# Functions, Rational Expressions & Complex Numbers

ClipSAT Printable Practice Suite (clipsat.org) • Topic 8.01 • 20 FRQs + 25 MCQs • Difficulty Progression • Cross-Lesson Synthesis

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** [Foundational] • [FRQ] • Target: Simplifying Rational Expressions & Domain Restrictions

Factor the numerator and denominator completely, state all domain restrictions, and simplify $\frac{2x^2 - 5x - 12}{x^2 - 16}$.

**2** [Foundational] • [FRQ] • Target: Adding & Subtracting Rational Expressions

Combine over a common denominator and simplify into a single reduced fraction: $\frac{3}{x-2} - \frac{2x-1}{x^2-4}$.

**3** [Foundational] • [FRQ] • Target: Solving Rational Equations

Solve the rational equation $\frac{4}{x-1} + \frac{2}{x+3} = 1$ for all valid real solutions.

**4** [Foundational] • [FRQ] • Target: Complex Number Arithmetic

Perform each complex number operation and write the result in standard form $a + bi$:

**a** $(5 - 3i)(2 + 4i)$

**b** $(3 - 2i)^2 - (4 + 5i)$

**5** [Foundational] • [FRQ] • Target: Complex Conjugates & Division

Divide the complex numbers $\frac{14 - 5i}{3 + 2i}$ by multiplying numerator and denominator by the complex conjugate, and express the result in the form $a + bi$.

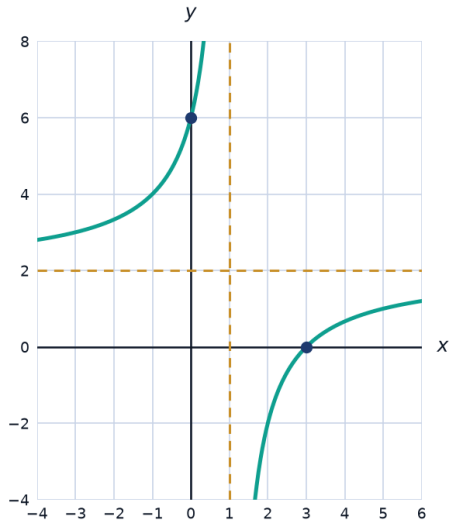
**6** [Foundational] • [FRQ] • Target: Powers of the Imaginary Unit $i$

Simplify $i^{45} + i^{102} - i^{79}$ using the periodicity of powers of $i$ ($i^4 = 1$).

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**7** [Foundational] • [FRQ] • Target: Asymptotes & Intercepts of Rational Functions

State the vertical asymptote, horizontal asymptote, $x$-intercept, and $y$-intercept of the rational function $f(x) = \frac{2x-6}{x-1}$ graphed below.




## Part I — Free-Response Questions (FRQs): Intermediate Modeling (8–14)

**8** [Intermediate] • [FRQ] • Target: Extraneous Solutions in Rational Equations

Solve the rational equation $\frac{x}{x-3} - \frac{3}{x-3} = 2$ and explain why it has no solution.

**9** [Intermediate] • [FRQ] • Target: Partial Fraction Decomposition

Find the constants $A$ and $B$ in the partial-fraction identity $\frac{5x+1}{(x-1)(x+2)} = \frac{A}{x-1} + \frac{B}{x+2}$.

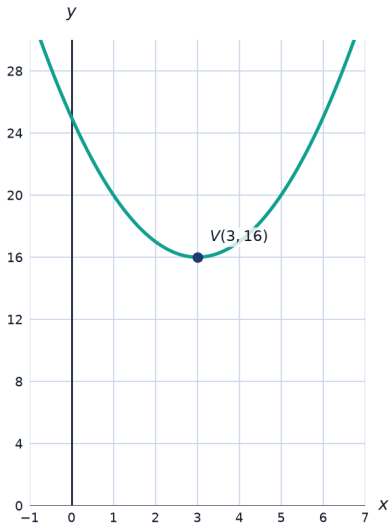
**10** [Intermediate] • [FRQ] • Target: Simplifying Complex Fractions

Simplify the complex fraction $\frac{\frac{1}{x}-\frac{1}{3}}{x-3}$ for $x \neq 0, 3$, and evaluate the simplified expression at $x = 3$.

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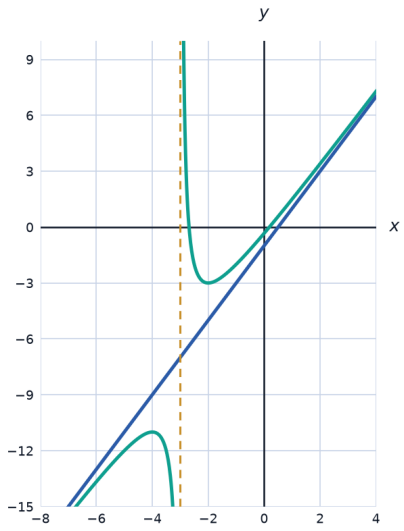
**11** [Intermediate] • [FRQ] • Target: Complex Roots of Quadratic Equations

Find the complex conjugate solutions to $x^2 - 6x + 25 = 0$ in the form $a \pm bi$, and verify that their sum is $6$ and their product is $25$.




**12** [Intermediate] • [FRQ] • Target: Quotient-Remainder Form of Rational Functions

Use polynomial long division to rewrite $\frac{2x^2 + 5x - 1}{x + 3}$ in the form $ax + b + \frac{r}{x + 3}$.



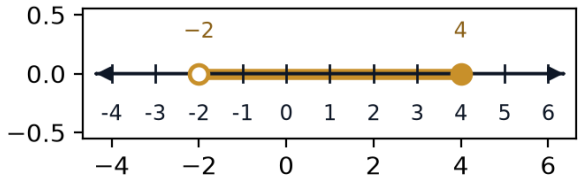

**13** [Intermediate] • [FRQ] • Target: Harmonic / Parallel Resistance Formulas

Two resistors $R_1$ and $R_2$ connected in parallel have equivalent resistance $R_{\text{eq}}$ given by $\frac{1}{R_{\text{eq}}} = \frac{1}{R_1} + \frac{1}{R_2}$. Solve for $R_2$ in terms of $R_{\text{eq}}$ and $R_1$, and compute $R_2$ when $R_{\text{eq}} = 12\,\Omega$ and $R_1 = 20\,\Omega$.

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**14** [Intermediate] • [FRQ] • Target: Solving Rational Inequalities

Solve the rational inequality $\frac{x-4}{x+2} \leq 0$ and express the solution set in interval notation.



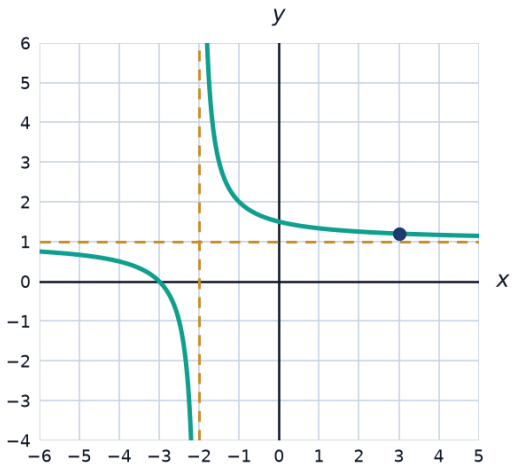

## Part I — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

**15** [Challenging] • [FRQ] • Target: Complex Conjugate Root Theorem • Cross-Lesson: Topic 4.01 (Cubic Polynomials)

A polynomial $P(x)$ of degree $3$ with real coefficients has zeros $x = 2$ and $x = 1 + 3i$, and satisfies $P(0) = -40$. Find the factored and expanded forms of $P(x)$.

**16** [Challenging] • [FRQ] • Target: Removable Discontinuities vs. Vertical Asymptotes • Cross-Lesson: Topic 3.01 (Quadratic Factoring)

Find the domain, removable discontinuity (hole) coordinates, and vertical asymptote of $g(x) = \frac{x^2 - 9}{x^2 - x - 6}$.




**17** [Challenging] • [FRQ] • Target: Upstream-Downstream Rational Modeling • Cross-Lesson: Topic 5.01 (Rates) & Topic 3.01 (Quadratics)

A kayaker paddles $12$ km upstream and $12$ km back downstream in a river whose current flows at $3$ km/h. If the total round-trip time is $3$ hours, set up a rational equation to find the kayaker's speed $v$ in still water.

**18** [Challenging] • [FRQ] • Target: Inverting Linear Fractional Transformations • Cross-Lesson: Topic 1.01 (Isolating Variables)

For a one-to-one rational function $f(x) = \frac{3x + 2}{x - 4}$ $(x \neq 4)$, find the formula for its inverse function $f^{-1}(x)$ and state the domain and range of $f$.

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**19** [Challenging] • [FRQ] • Target: Modulus Algebra in the Complex Plane • Cross-Lesson: Topic 7.01 (Pythagorean Theorem)

In the complex plane, find the modulus $|z|$ of $z = \frac{(3+4i)(1-i)}{2-2i}$ without expanding the numerator.

**20** [Challenging] • [FRQ] • Target: Rational-Reciprocal Identities & Optimization • Cross-Lesson: Topic 3.01 (Completing the Square)

If $x > 0$ and $x + \frac{4}{x} = 6$, find the exact value of $x^2 + \frac{16}{x^2}$ and the minimum possible value of $t + \frac{4}{t}$ for all $t > 0$.

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## Part II — Multiple-Choice Questions (MCQs): Foundational Concepts (21–28)

**21** [Foundational] • [MCQ] • Target: Simplifying Rational Expressions

Given the algebraic relationship in which for $x \neq 3$, which expression is equivalent to $\frac{x^2-9}{x-3}$?

**a.** $x - 3$

**b.** $x + 3$

**c.** $x^2 + 3$

**d.** $x - 9$

**22** [Foundational] • [MCQ] • Target: Adding Complex Numbers

If $i = \sqrt{-1}$, what is the sum $(7 - 4i) + (-2 + 9i)$?

**a.** $5 + 5i$

**b.** $5 - 13i$

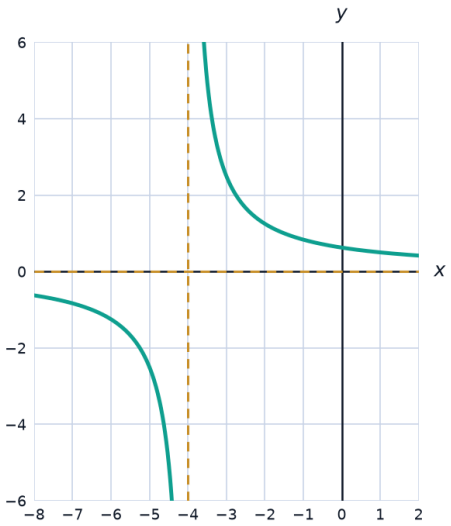
**c.** $9 + 5i$

**d.** $9 - 5i$

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**23** [Foundational] • [MCQ] • Target: Vertical Asymptotes

What is the vertical asymptote of the graph of $f(x) = \frac{5}{2x+8}$?




**a.** $x = -4$

**b.** $x = 0$

**c.** $x = 4$

**d.** $x = 5$

**24** [Foundational] • [MCQ] • Target: Proportions & Rational Equations

Within the real coordinate domain, to satisfy the synthesis condition, if $\frac{6}{x-2} = \frac{3}{4}$, what is the value of $x$?

**a.** 6

**b.** 8

**c.** 10

**d.** 12

**25** [Foundational] • [MCQ] • Target: Product of Complex Conjugates

What is the product $(4 + 3i)(4 - 3i)$?

**a.** 7

**b.** $16 - 9i$

**c.** 25

**d.** $7 + 24i$

**26** [Foundational] • [MCQ] • Target: Adding Fractions with Variable Denominators

Combine the rational terms $\frac{2}{x} + \frac{5}{3x}$ $(x \neq 0)$ into a single simplified fraction.

**a.** $\frac{7}{4x}$

**b.** $\frac{7}{3x}$

**c.** $\frac{11}{3x}$

**d.** $\frac{10}{3x^2}$

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**27** [Foundational] • [MCQ] • Target: Powers of $i$

What is the value of $i^{26}$?

**a.** $-1$

**b.** $1$

**c.** $-i$

**d.** $i$

**28** [Foundational] • [MCQ] • Target: Domain of Rational Functions

For what value of $x$ is the function $h(x) = \frac{x+5}{x^2-6x+9}$ undefined?

**a.** $-5$

**b.** $-3$

**c.** $0$

**d.** $3$

## Part II — Multiple-Choice Questions (MCQs): Intermediate Exam Practice (29–37)

**29** [Intermediate] • [MCQ] • Target: Rewriting Improper Rational Expressions

Rewrite the improper rational function $\frac{3x+7}{x+2}$ $(x \neq -2)$ in quotient-plus-remainder form.

**a.** $3 + \frac{1}{x+2}$

**b.** $3 + \frac{7}{x+2}$

**c.** $3 + \frac{5}{x+2}$

**d.** $3 - \frac{1}{x+2}$

**30** [Intermediate] • [MCQ] • Target: Complex Division & Real/Imaginary Parts

Analyze the configuration in which when $\frac{10}{1-2i}$ is written in the form $a + bi$, what is the value of $a + b$?

**a.** $2$

**b.** $4$

**c.** $6$

**d.** $8$

**31** [Intermediate] • [MCQ] • Target: Filtering Extraneous Roots

What are all real solutions to $\frac{x^2}{x-4} = \frac{16}{x-4}$?

**a.** $x = 4$ only

**b.** $x = -4$ only

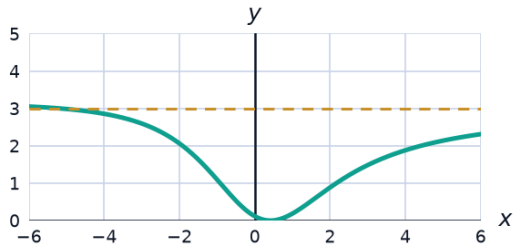
**c.** $x = 4$ and $x = -4$

**d.** No real solution

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**32** [Intermediate] • [MCQ] • Target: Horizontal Asymptotes of Equal-Degree Rationals

What is the horizontal asymptote of $f(x) = \frac{6x^2 - 5x + 1}{2x^2 + 9}$?




**a.** $y = 0$

**b.** $y = \frac{1}{3}$

**c.** $y = 3$

**d.** $y = 6$

**33** [Intermediate] • [MCQ] • Target: Partial Fractions Cover-Up Method

If $\frac{4x - 2}{x^2 - 1} = \frac{A}{x - 1} + \frac{B}{x + 1}$ for all $x \neq \pm 1$, what is the value of $A$?

**a.** 1

**b.** 2

**c.** 3

**d.** 4

**34** [Intermediate] • [MCQ] • Target: Simplifying Compound Fractions

Simplify the compound reciprocal fraction $\frac{\frac{1}{a} + \frac{1}{b}}{\frac{1}{ab}}$ for positive real numbers $a, b > 0$.

**a.** $\frac{1}{a + b}$

**b.** $a + b$

**c.** $ab$

**d.** $\frac{a + b}{a^2 b^2}$

**35** [Intermediate] • [MCQ] • Target: Equating Real & Imaginary Parts

If $(x + yi)(2 - i) = 8 + i$, where $x$ and $y$ are real numbers, what is the value of $x + y$?

**a.** 3

**b.** 4

**c.** 5

**d.** 6

**36** [Intermediate] • [MCQ] • Target: Rational Expressions for Combined Work Rates

Printer A can print a batch of reports in $x$ hours, and Printer B can print the same batch in $x + 3$ hours.

Which expression represents the fraction of the batch they can print together in 1 hour?

**a.** $\frac{2}{2x + 3}$

**b.** $\frac{2x + 3}{x(x + 3)}$

**c.** $\frac{x(x + 3)}{2x + 3}$

**d.** $\frac{1}{x(x + 3)}$

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**37** [Intermediate] • [MCQ] • Target: Reconstructing Quadratics from Complex Conjugate Roots

Which of the following quadratic equations has roots $2 + i$ and $2 - i$?

**a.** $x^2 - 4x + 3 = 0$

**b.** $x^2 - 4x + 5 = 0$

**c.** $x^2 + 4x + 5 = 0$

**d.** $x^2 - 2x + 5 = 0$

## Part II — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • [MCQ] • Target: Periodic Series of Powers of $i$ • Cross-Lesson: Topic 4.01 (Exponents & Periodicity)

What is the value of the sum $i^1 + i^2 + i^3 + \ldots + i^{99}$?

**a.** $-1$

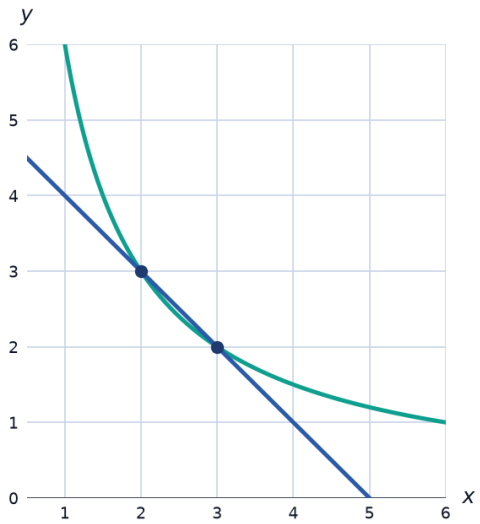
**b.** $0$

**c.** $i$

**d.** $-i$

**39** [Challenging] • [MCQ] • Target: Hyperbola-Line Intersection System • Cross-Lesson: Topic 2.01 (Systems) & Topic 3.01 (Quadratics)

In the $xy$-plane, at how many points does the rational curve $y = \frac{6}{x}$ intersect the line $x + y = 5$?




**a.** Zero

**b.** One

**c.** Two

**d.** Three

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**40** [Challenging] • [MCQ] • Target: Rational Dilution Modeling & Asymptotic Behavior • Cross-Lesson: Topic 5.01 (Mixture & Concentration)

A chemistry student adds $x$ liters of pure water to $20$ liters of a $40\%$ acid solution. Which rational function gives the new acid concentration $C(x)$ as a decimal, and what is $\lim_{x\to\infty} C(x)$?

**a.** $C(x) = \frac{8}{20+x}$, horizontal asymptote $C = 0$

**b.** $C(x) = \frac{8+x}{20+x}$, horizontal asymptote $C = 1$

**c.** $C(x) = \frac{0.40}{20+x}$, horizontal asymptote $C = 0$

**d.** $C(x) = \frac{20}{8+x}$, horizontal asymptote $C = 0$

**41** [Challenging] • [MCQ] • Target: Integer Divisibility via Rational Division • Cross-Lesson: Topic 4.01 (Polynomial Division)

For how many integer values of $n$ is $\frac{2n+15}{n+2}$ also an integer?

**a.** $2$

**b.** $4$

**c.** $6$

**d.** $8$

**42** [Challenging] • [MCQ] • Target: 90-Degree Rotation by $i$ in the Complex Plane • Cross-Lesson: Topic 7.01 (45-45-90 Right Triangles)

If $z = 3 + 4i$, what is the distance in the complex plane between $z$ and $iz$?

**a.** $5$

**b.** $5\sqrt{2}$

**c.** $7$

**d.** $10$

**43** [Challenging] • [MCQ] • Target: Non-Zero Right-Hand-Side Rational Inequalities • Cross-Lesson: Topic 2.01 (Inequalities)

What is the solution set to the rational inequality $\frac{x+5}{x-2} \geq 2$?

**a.** $(2, 9]$

**b.** $[9, \infty)$

**c.** $(-\infty, 2) \cup [9, \infty)$

**d.** $[-5, 2)$

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**44** [Challenging] • [MCQ] • Target: Inverse Pythagorean Theorem for Right-Triangle Altitude • Cross-Lesson: Topic 7.01 (Right-Triangle Altitude)

A right triangle has legs of lengths $\frac{1}{a}$ and $\frac{1}{b}$, where $a, b > 0$. Which rational expression gives the length of the altitude drawn to the hypotenuse?

**a.** $\frac{1}{a+b}$

**b.** $\frac{1}{\sqrt{a^2+b^2}}$

**c.** $\frac{ab}{\sqrt{a^2+b^2}}$

**d.** $\frac{\sqrt{a^2+b^2}}{ab}$

**45** [Challenging] • [MCQ] • Target: Iterated Composition of Rational Functions • Cross-Lesson: Topic 4.01 (Function Composition)

If $f(x) = \frac{x-1}{x+1}$, what is $f(f(x))$ for all $x \neq -1, 0$?

**a.** $x$

**b.** $-\frac{1}{x}$

**c.** $\frac{1}{x}$

**d.** $-x$

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